

Texas A&M mini MPL Operation SOP

Operator Name: _____ Operation Date: _____

1) Start lidar measurement

- a) Connect the lidar power cables and data cables ☐
- b) Turn on the lidar power rocker switch ☐
- c) Unscrew the laser cover ☐
- d) Turn on the key and the red rocker switch to turn on the laser ☐
- e) Do not turn off laser (Both key and switch) while the MPL software is collecting data.
- f) Open sigmaMPL software if not automatically opened, open File -> Real time hardware control ☐
- g) Software may automatically start collecting. Check config files and resolution, stop if need to change. ☐
- h) In configuration, record Lidar Afterpulse file: _____
- i) Record Lidar Overlap file: _____
- j) Record Deadtime file: _____
- k) Record Lidar temporal resolution: _____ and vertical resolution: _____
- l) Click Save Data, make sure the entry box on the right side is empty ☐
- m) Click "Start Collecting". Record Lidar Laptop Time: _____
- n) Wait for the first green bar to finish, and record the file name: _____

2) During lidar measurement

- a) Monitor Raw, NRB, Depol, and Hardware tab and record field notes:

- b) If the laser temperature reaches 35 °C, stop collecting and turn off the laser

3) Shut down procedure

- a) Click stop collection ☐
- b) Close software ☐
- c) Turn off the laser (key and then the red rocker switch). Make sure laser is off. ☐
- d) Flip rocker switch off ☐
- e) Screw the cover back in and unplug cables. ☐